Proprietary to General Electric Company
Direction 5438996-8EN, **Rev#** 9
Brivo CT385 Series Service Methods (Adv)
Module Version 2.0, **Version Date** 1/9/14

Gantry Balance Procedure

(Click Here for Summary Version)

1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedure	Finalization
1	Not Applicable	1 - 2	0.5 hour

2 Overview

This procedure should be performed whenever the rotating Gantry is serviced. It is intended to 1) check or create sensitivity matrix, and 2) balance the Gantry in both static and dynamic vectors.

Last Revised:	14 November, 2013

3 Preliminary Requirements

3.1 Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacturer
Torque wrench kit (Must order from the tool depot. The field torque kit does not have the capacity required for the balance weight fasteners.)	1	-	46-268445G1 or equivilent	-
Deep-well socket drive	1	-	Part of Balance Kit shipped with system	-
Spring balance	1	-	U0012BF shipped with system	-
torque wrench	1	-	46-268521G1 or equivalent	-

3.2 Safety



DANGER

CRUSH HAZARD.
UNEXPECTED GANTRY ROTATION CAN CAUSE SERIOUS INJURY.
ALWAYS DISABLE AXIAL DRIVE BEFORE TOUCHING ROTATING ASSEMBLY. KEEP AWAY FROM ROTATING ASSEMBLY WITH AXIAL DRIVE ENABLED.



WARNING

IMPACT HAZARD.
FAILURE TO TORQUE TRIM WEIGHT FASTENING HARDWARE WILL RESULT IN DAMAGING PROJECTILE EVENTS THAT WILL RESULT IN SEVERE EQUIPMENT DAMAGE AND POSSIBLE PERSONAL INJURY.
TORQUE ALL FASTENERS TO PROPER LEVELS.

4 Procedure

1. Position the Table at its lowest position.
2. Remove the gantry left side cover.

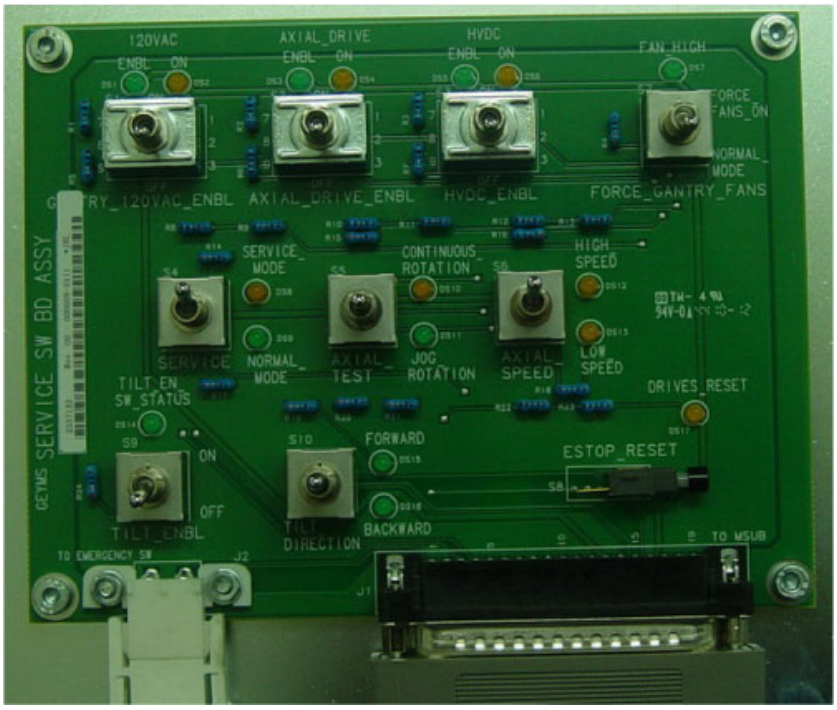


NOTICE

Always turn OFF the HVDC before the 120VAC. Turning OFF 120VAC power before HVDC power can result in equipment damage.

3. Disable Axial Drive, HVDC, and 120 VAC service switches from the service switch panel.

Illustration 1: Service Switch Board



4. Remove the gantry covers as required.
5. Enable the 120VAC, HVDC and Axial Drive service switches from the service switch panel in order to run the balance test.

Illustration 2: Gantry Balance Weight Locations

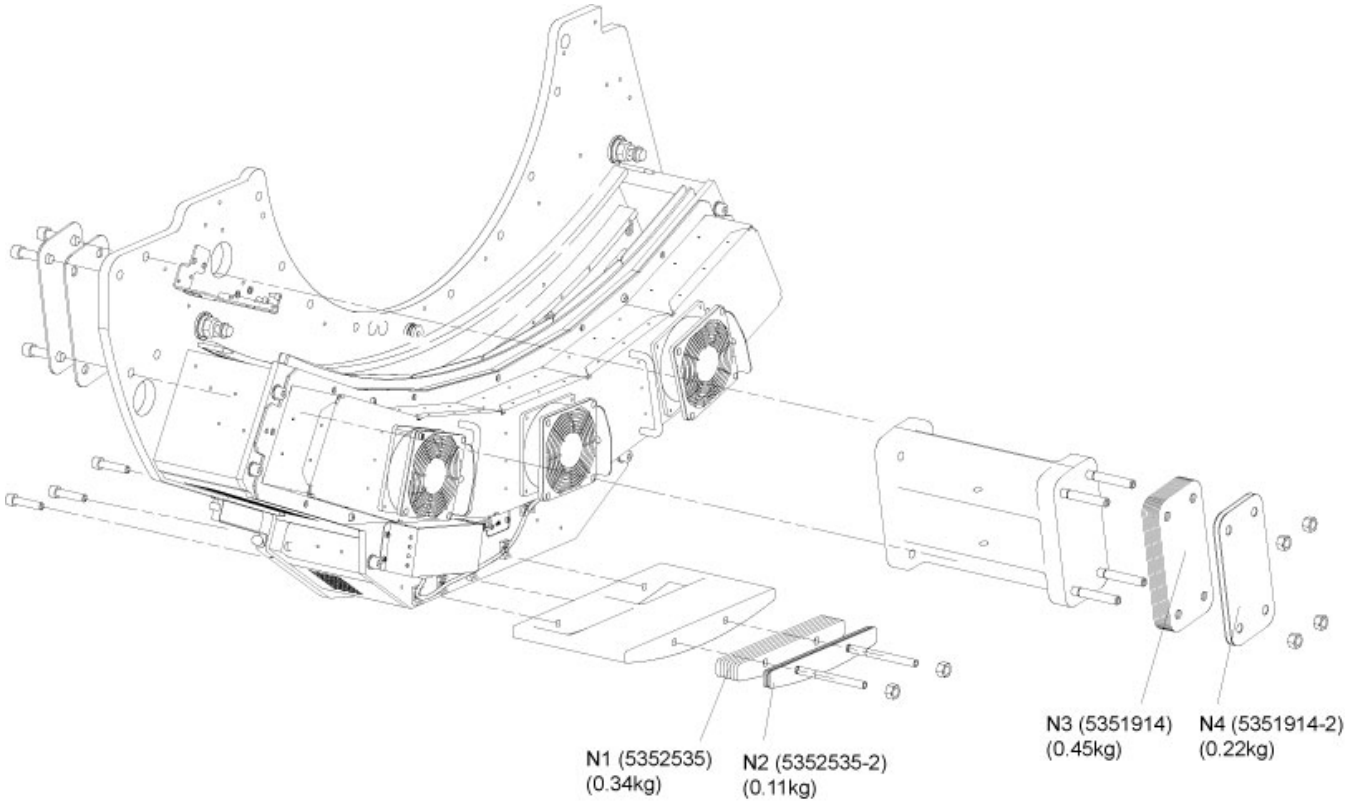


Table 1: Max Number of Gantry Balance Weight Adjuster Sheet

WEIGHT ADJUSTER SHEET	MAX NUMBER OF INSTALLED PCS
N1 (5352535)	15
N2 (5352535-2)	3
N3 (5351914)	12
N4 (5351914-2)	2

6. Perform the **Rough** weight balance check/adjustment by observing the free movement:
    a. Rotate the gantry until the main weight (under the DAS Assy) reaches the 6 o'clock position.
    b. Verify that the gantry rotates on its own, due to unbalance.

- If the Gantry rotates in the CW direction, increase the side weight.
- If the Gantry rotates in the CCW direction, decrease the side weight.

c. Repeat the above step until the Gantry is balanced.

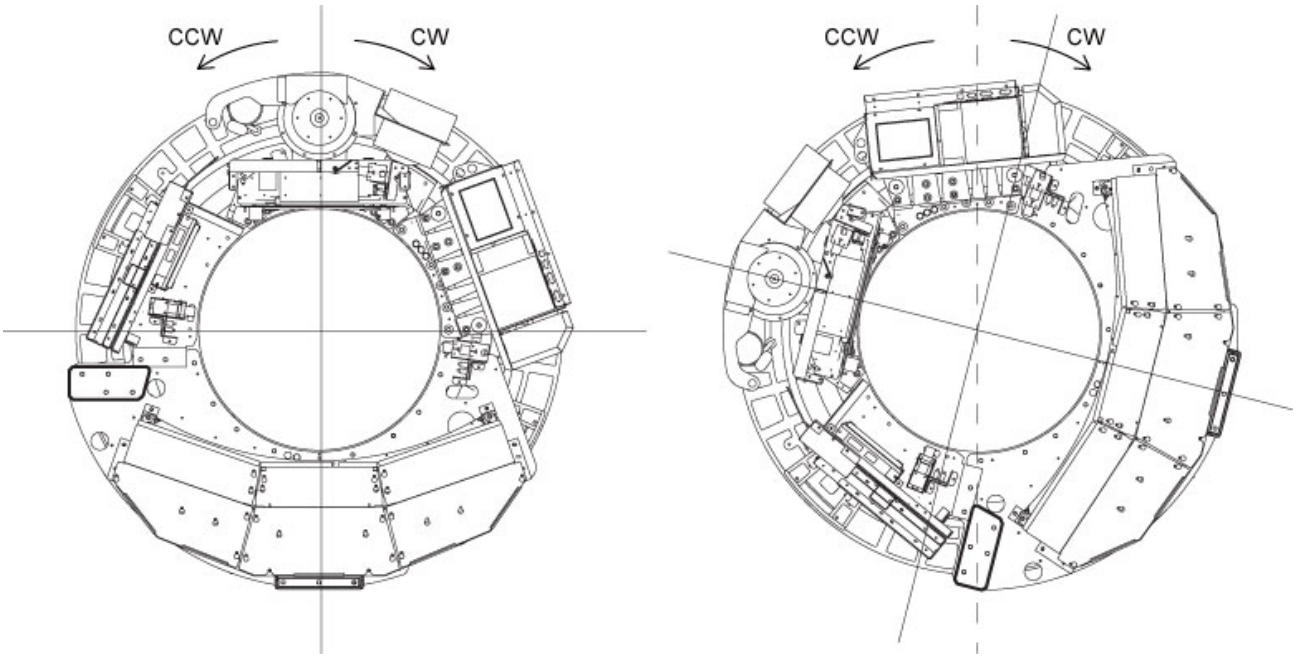
d. Rotate the gantry until the side weight reaches the 6 o'clock position.

e. Verify that the gantry rotates on its own, due to unbalance.

- If the Gantry rotates in the CW direction, decrease the main weight.
- If the Gantry rotates in the CCW direction, increase the main weight.

f. Repeat the above steps until the gantry is balanced.

Illustration 3: Rough Weight Balance



7. Perform the **Exact** weight balance check/adjustment using a spring balance:

a. Rotate the gantry until the tube reaches the 12 o'clock position.

b. Measure the force required (F1 and F2) along the tangential direction to initiate gantry rotation by pulling the gantry with a spring balance.

- If the value (F2-F1) is positive, decrease the weight in the side weight, until |F2-F1| is less than or equal to 0.5kg.
- If the value is negative, increase the weight in the side weight.

c. Rotate the gantry until the tube reaches the 3 o'clock position.

d. Measure the force required (F3 and F4) along the tangential direction to initiate gantry rotation by pulling the gantry with a spring balance.

- If the value (F4-F3) is positive, decrease the weight in the main weight, until |F4-F3| is less than or equal to 0.5kg.
- If the value is negative, increase the weight in the main weight.

Illustration 4: Gantry Static Balance Force Measurement

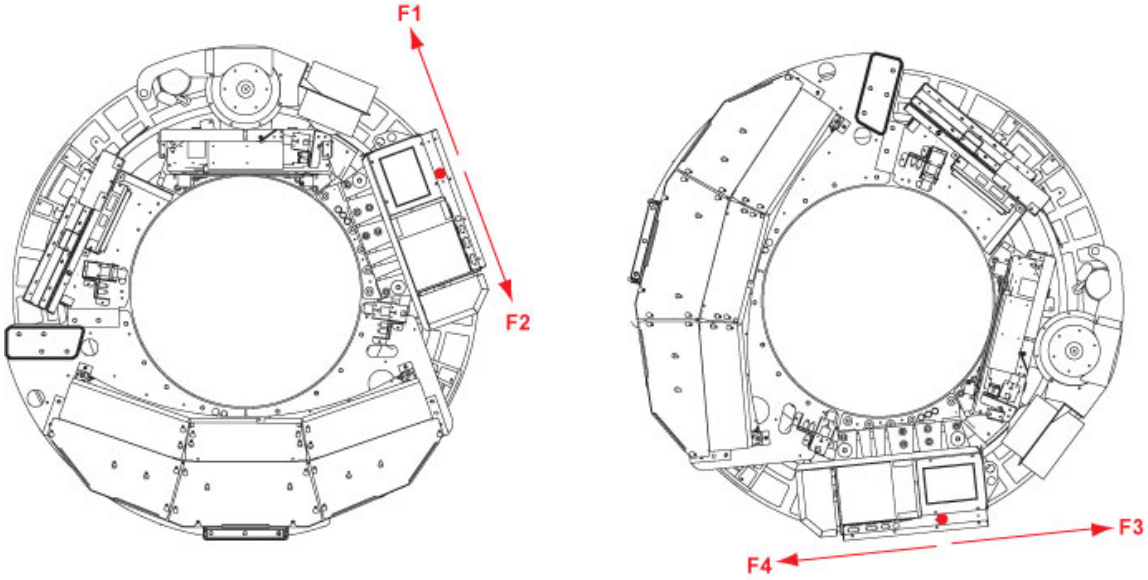
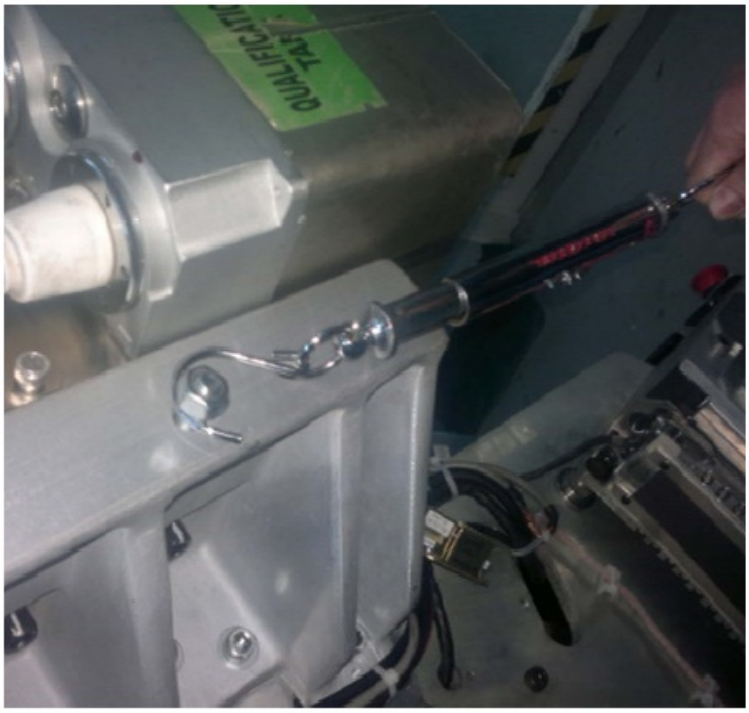


Illustration 5: Position of Spring Balance



NOTE: If there isn't a nut on the JEDI bracket for the spring balance, please use other substitutes.

5 Finalization

1. Disable the axial drive, HVDC and 120VAC service switches from the service switch panel.
2. Install the front cover (and rear if removed), top covers and right side cover.
3. Enable the 120VAC, HVDC and axial drive service switches from the service switch panel and install the gantry left side cover.

   Remember to enable the table drives using the push button on the bottom right of the service switch panel.
4. Perform FASTCAL. Ensure no pop-up windows occur and gesyslog is error free.

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